GENBE302 – FUNDAMENTAL OF ELECTRICITY AND ELECTRONIC

DATE: .../.../2025

PERIOD: .....

INSTITUT TECHNIQUE BAPTISTE (ITB TSS)
Study Success Discipline

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM II EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical technology

RQF LEVEL: 3

MODULE TITLE: **FUNDAMENTAL OF ELECTRICITY AND ELECTRONIC**

MARKS: **...../100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Twenty-four (24) questions distributed in Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: Between (11-17) questions, all are **Compulsory**.	**55 marks**
Section **B**: Five (5) questions, choose and answer any three (3).	**30 marks**
Section **C**: Two (2) questions, choose and answer any one (1)	**15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

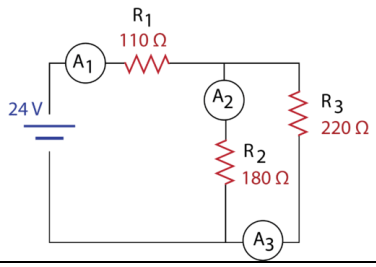
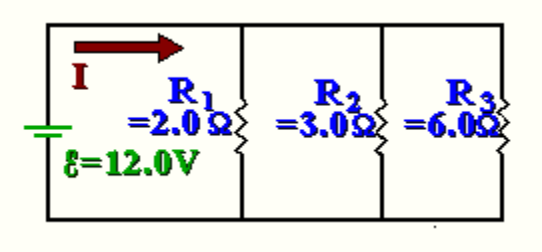
SECTION A: Attempt all questions **(55 marks)**

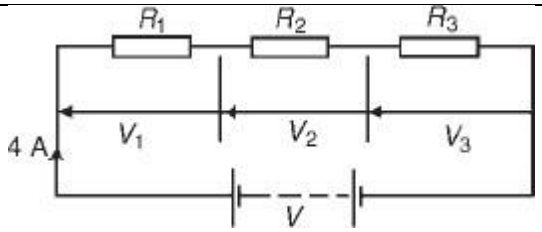
01.	**What is the unit of electric current?** a) Volt b) Ampere c) Ohm d) Watt	**(2Marks)**
02.	**Which of the following is true about Ohm's Law?** a) V=I×RV = I \times RV=I×R b) P=I×RP = I \times RP=I×R c) I=V×RI = V \times RI=V×R d) V=I+RV = I + RV=I+R	**(2Marks)**
03.		**(2Marks)**
04.	**What is the peak value of an AC voltage waveform with an RMS value of 10V?** a) 10V b) 14.14V c) 20V d) 7.07V	**(2Marks)**
05.	**Kirchhoff's Current Law (KCL) states that:** a) The sum of all voltages around a closed loop equals zero. b) The sum of currents entering a junction equals the sum of currents leaving. c) The sum of all powers in a circuit equals zero. d) The current is inversely proportional to the resistance.	**(2Marks)**
06.	**Which theorem states that the maximum power is transferred when the load resistance equals the source resistance?** a) Thevenin's Theorem b) Norton's Theorem c) Superposition Theorem d) Maximum Power Transfer Theorem	**(2Marks)**
07.	Describe the function of a diode. Give one practical application	**(2Marks)**
08.	**What is the unit of power in an electrical circuit?**	**(2Marks)**

	a) Ohm b) Volt c) Ampere d) Watt	
09.	Explain how a transistor can be used as a switch	**(2Marks)**
10.	Discuss the role of resistors in controlling current flow.	**(2Marks)**
11.	**True or False** **The unit of electric current is Ampere.**	**(3Marks)**
12.	What happens to the total resistance when more resistors are added in parallel?	**(3Marks)**
13.	**True or False** **Kirchhoff's Voltage Law states that the sum of all currents in a circuit must be equal to zero.**	**(3Marks)**
14.	Explain the difference between conductors, semiconductors, and insulators with examples	**(3Marks)**
15.	Three resistors, $R_1 = 4\ \Omega$, $R_2 = 6\ \Omega$, and $R_3 = 12\ \Omega$, are connected in parallel. Find the total resistance.	**(3Marks)**
16.	**Match the term to the correct definition.** **Term** — **Definition** 1. **Ohm's Law** — a) The sum of voltages around a loop equals zero. 2. **Kirchhoff's Voltage Law** — b) Relationship between voltage, current, and resistance in a circuit. 3. **Peak Value** — c) The maximum value of an AC waveform. 5. **Thevenin's Theorem** — e) Replaces a complex network with a single voltage source and series resistance.	**(10Marks)**
17.	Match electrical quantities with their definitions in a table below	**(10Marks)**

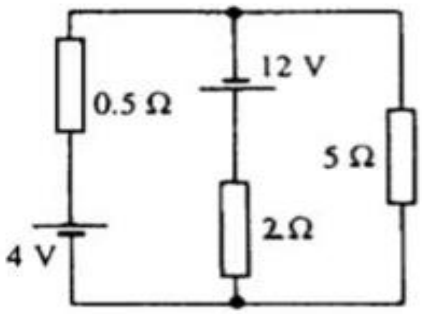
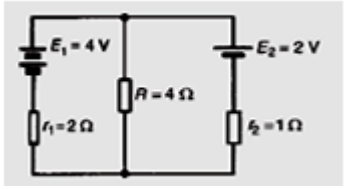
	ANSWERS ELECTRICAL	ANSWERS ELECTRICAL	ANSWERS ELECTRICAL		
	1……	1. Power	A. describes the "pressure" that pushes electricity		
	2……	2. Energy	B. refers to the flow of electricity in an electronic circuit, and to the amount of electricity flowing through a circuit.		
	3……	3. Voltage	C. is a measure of the opposition to current flow in an electrical circuit.		
			D. is the power an atom's charged particles have to cause an action or move an object.		
			E. measures the rate of electrical energy transfer by an electric circuit per unit of time.		

Section B: Attempt any three (3) questions **(30 marks)**

18.	In the combination circuit sketched below, find: a) The equivalent resistance for the circuit. b) The total current through the circuit. The current through each individual resistor 	**(10marks)**
19.	In the following schematic diagram, find the total current (I). 	**(10marks)**
20.	A 100 W electric light bulb is connected to a 250 V supply. Determine (a) the current flowing in the bulb, and (b) the resistance of the bulb	**(10marks)**
21.	Explain The main characteristics of a series circuit	**(10marks)**
22.	For the circuit shown in Figure below, determine (a) the battery voltage V, (b) the total resistance of the circuit, and (c) the values of resistance of resistors R1, R2 and R3, given that the p.d.'s across R1, R2 and R3 are 5 V, 2 V and 6 V respectively.	**(10marks)**

	 R_1 R_2 R_3 4 A V_1 V_2 V_3 V	

Section C: Attempt only one (1) question **(15 marks)**

23.	Determine the current in the 5Ω resistance of the network shown in Figure below using Norton's theorem and Thevenin's theorem. Hence find the currents flowing in the other two branches 	**(15marks)**
24.	Figure below shows a circuit containing two sources of e.m.f., each with their internal resistance. Determine the current in each branch of the network by using the superposition theorem. 	**(15marks)**

END OF ASSESSMENT